Name

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Handout 4 · Code clinic pack

The four-step routine

Step	What to do
1 · Read	Read the LAST line of the error first: its type and its message.
2 · Locate	Find the arrow ----> and the line number it points to.
3 · Print	Print the variables on that line: print(img.shape), print(type(x)).
4 · Search	Search the exact message, or ask an AI assistant to explain it (not to fix it).

Still stuck after 15 minutes? Explain the problem out loud to your pair — or a rubber duck.

Five errors you will meet this term

Error	Message you see	What it usually means
FileNotFoundError	No such file or directory: 'my_photo.jpg'	File not uploaded, or name misspelled (names are case-sensitive)
TypeError	'str' object is not callable	You reused a function's name as a variable
IndexError	index 400 is out of bounds for axis 0 with size 300	You asked for a row or column that doesn't exist
ValueError	operands could not be broadcast together with shapes (300,451,3) (200,300,3)	Combining two images of different sizes — resize first
No error at all!	250 + 10 gives 4 in a uint8 image	Values wrap past 255 — convert to int and use np.clip

Broken-notebook challenge

Open **Week02_Tutorial_Broken_Notebook.ipynb**. Five cells, five bugs. For each: run it, apply the routine, fix it, and note the fix below.

Bug	Error type (or symptom)	Cause	My fix
1	In this cell, the image doesn’t load properly and invokes a FileNotFoundError	The filename provided to load_image() is “Photo.JPG” when the filename is “Photo.jpg”	We supply load_image() with the right filename “Photo.jpg”

Bug	Error type (or symptom)	Cause	My fix
2	IndentationError: there is a missing indentation somewhere	The “return 255 - img” is not indented	Add an indentation to that line
3	show() is said to be not a function (TypeError)	The “show” function name is overwritten by the new image’s variable name called “show”	Rename the new image’s variable name to a more appropriate one, “swapped”
4	IndexError: The value at index 300 cannot be read	Since arrays are indexed starting from 0, then the highest index would be 299	For a more reliable approach, use negative indexing to always get the very bottom index, then change “10” to “0” to get the leftmost index
5	There are dark spots across the brightened image	Since RGB values can only be between 0 and 255, values that go beyond 255 cannot be read properly	Allow the RGB values to go above 255 by converting them into a 16-bit integer, limit them back to 255 by using np.clip(), then convert them back into an 8-bit integer

My bug diary

Log every bug that takes you more than 10 minutes. By Week 14 this shows how far you have come.

Date	What I was doing	Error type + last line	Cause	Fix	Time

My AI-assistant statement

I will use AI assistants to **explain** errors or give hints, not to write my work. I will be able to explain every line of code I submit, and I will record any AI help in my R&D journal.

Signed: Vince Matthew Caballero Date: November 5, 2026